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Corporate Physical Security Analytic Sector Report

Executive Summary

This sector report will review the history of corporate physical security automation from the initial need that rose during the industrial revolution, to the expansion of what a security team member is expected to do today, and what the future could look like for the profession. The use of AI has the potential to upskill a traditionally low-level role and "mitigate the need for humans to perform tasks that fall into the classic "3Ds" of automation: dirty, dangerous, and dull" (Loyear, & Slotnick, 2025). Major concerns over data privacy and reasonable data retention timelines remain at the forefront. Security teams implementing AI with privacy in mind, working cross-functionally with legal and privacy teams, have the potential to radically disrupt archaic processes into a powerhouse of proactive functions and improve industry perception.

History

At the start of the Industrial Revolution, working-aged people were pushed towards urban centers to earn a living in industry (Jack, 2025). This introduced a need not only for security personnel, but also for a way to track return on investment (ROI) and prevent incidents through security patrolling. This sparked an innovation that led to the creation of the first watchman's clock in 1855 (Reid, 2024). This was the main form of patrolling until the 1980's, highlighting how the sector "is notoriously slow-moving when it comes to tech innovation" (Russell, 2025). Access control features began to appear in the 1930's with electric key systems, moved to magnetic stripe cards in the 1960's, and RFID, which is still used today, emerged in the 1980s (Santiago, 2025). The use of Closed-Circuit Television (CCTV), a major aspect of physical security tooling, emerged in the 1950's and expanded usage in the 1960's. (Williams, 2003).

Current Trends

Today, physical security has evolved beyond deterrence by presence and personnel surveillance to utilizing multiple forms of technology. Guards now have access to apps on cellular devices to complete patrols, record video and audio, access alarm alerts, receive real-time updates, request backup, and create incident reports. Security robots, launched by Cobalt AI, are now also equipped with the ability to respond to alarms, detect hazards, request backup, and interact with individuals requesting assistance through video. What used to require multiple personnel can now be accomplished by a singular physical guard and a security robot. "Video surveillance systems are increasingly integrated with other security controls such as access controls, and intruder and fire detection systems. Integration with access controls enables synchronized monitoring of physical access points and video footage to show incidents such as a door being forced open, and to tie incidents to a specific person" (Howarth, 2024). Yet with all the new tools available, "only 6 percent of organizations have fully embedded automation into their operations with continuous optimization" (Barr, 2026). This may indicate the resistance to change in the security industry and the difficulty of incorporating AI into legacy systems due to budget constraints. Areas where AI remains underutilized and could have the greatest impact include pattern detection within incident reports, alarm data, and unusual activity captured on CCTV, helping identify risk and move the industry from reactive to proactive operations.

Future Growth

"Surveillance camera systems and physical and logical access control and monitoring systems produce vast amounts of data. More data, in fact, than could ever be monitored and analyzed by human security teams. Analyzing mountains of data to achieve actionable outcomes is exactly

the scenario in which current AI applications excel" (Briscoe, 2025). Those actionable outcomes could include predicting high incident areas, assessing risk in new markets, providing trend findings to stakeholders to showcase ROI, assisting in identifying roadmap strategies, and reducing the risk and harm of security team members through other means of patrolling. This opens the opportunity to upskill guards into higher-level tasks, overseeing and validating AI findings for accuracy, and providing new career paths. Loyear and Slotnick call out several possible new roles, such as AI surveillance analyst, security data interpreter, Human-machine liaison, security automation engineer, and more. Each of these roles involves ensuring a human stays in the loop to prevent blind acceptance of AI analyses. "The machines may see the pattern, but human operators apply perspective, intuition, and action." (2025).

Potential Costs and Risks

With the rise in AI, there is not only potential for new roles, but also the potential for job loss as well, particularly by those who are resistant to change and do not take the time to learn new skills. The robotic guarding system reduces business costs, as they do not need breaks, benefits, or pay. What once needed 3 personnel for 24 hours needs only one robot. Risks posed by this new guarding option include the need to verify decisions made when responding to an alarm, reviewing incident data for accuracy, and malware attacks that could result in visual or physical access breaches. Environmental risks include increased power consumption as many systems operate 24/7 and hardware churn (Howarth, 2024), through retrofitting devices that are not compatible with the new technology. Bias in training data from operators could lead to bypassing an alarm through false negatives or responding to false positives (Lawlor, 2021), resulting in unreported damages or a major access control breach. Using third-party software raises the question of data privacy, risking data being retained longer than acceptable by GDPR or other regulatory standards, and the potential of that third party utilizing the data to train other LLM's without proper safeguards in place.

Conclusions and Recommendations

Utilizing AI can have the power to elevate the security profession; those who do not train their teams to upskill risk falling behind or being eliminated from the industry. The ability to surveil at a massive scale requires a framework built on ethics and privacy practices to ensure regulatory adherence and industry respect while sorting through an ever-growing amount of data. Utilizing AI in patrols, access control, and CCTV through robotics provides security teams with the edge to move from a cost center to becoming a major business asset (Briscoe, 2025). New access control tools merging video and access control to detect behavioral attacks, such as Human-Centric Zero Trust Identity Architecture (HC-Zita) (Nsoh, 2026), and the No Pain Device, designed to detect incidents of aggression through speech recognition (Giorgio, 2024), will become integral in preventing acts of workplace violence or armed shooters through alerting security team members not to issue visitor credentials or locking down access controls when a risk threshold is reached. Robotic guarding, AI video and document analysis, and tools that can detect bad actors through tone or body language and shut down access control are all positive outcomes for AI usage. There is major potential for positive new paths within the security industry with AI, provided they build with privacy in mind, work cross-functionally with legal to ensure secure contracts on third-party software usage, and keep a human in the process.

References

Barr, J. (2026). *State of security automation.* Faulkner Information Services. https://faulkner-com.eu1.proxy.openathens.net/securitymgmt/state-of-security-automation/

Briscoe, S. (2025). *Understanding the evolving role of security: 2025 security trends research.* ASIS International. https://www.asisonline.org/globalassets/publicationsand-resources/asis-2025-security-trends-research-report.pdf

Giorgio, A. (2024). No pain device: Empowering personal safety with an artificial intelligence-based nonviolence embedded system. *Electronics*, *13*(9), 1766. https://doi.org/10.3390/electronics13091766

Howarth, F. (2024). *Video surveillance trends*. Faulkner Information Services. https://faulkner-com.eu1.proxy.openathens.net/securitymgmt/video-surveillance-trends-2/

Jack. (2025). *Watch-clocks: 120 years of timed security.* The Security Museum. https://thesecuritymuseum.com/2025/11/04/watch-clocks-120-years-of-timed-security/

Lawlor, S. (2021). Where we stand with AI. *Security Management*. https://www.asisonline.org/security-management-magazine/monthly-issues/security-technology/archive/2021/april/where-we-stand-with-ai/

Loyear, R., & Slotnick, J. (2025). From guards to guardians: Reimagining security work in the age of AI and robotics. *Security Management.* https://www.asisonline.org/security-management-magazine/articles/2025/08/guards-to-guardians/?_t_id=8yEa3b8FuoYiSDOGiKOD8A%3d%3d&_t_uuid=rKPWALEQSHyqdiceR7CvQQ&_t_q=AI+&_t_tags=language%3aen%2csiteid%3ab1140b07-9e31-4808-809a-878911c7f3f1%2candquerymatch&_t_hit.id=ASIS_Models_Pages_SMArticleDetailPage/_4e015e3b-c8ad-49be-b4de-7289042a6988_en&_t_hit.pos=14

Nsoh, J. T. (2026). Human-centric zero trust identity architecture for the fifth industrial revolution: A JEPA-driven approach to adaptive identity governance. *Electronics*, *15*(9), 1878. https://doi.org/10.3390/electronics15091878

Reid, P. (2024). *A solution for unreliable employees.* McMinn County Living Heritage Museum. https://www.livingheritagemuseum.org/2024/05/30/12738/a-solution-for-unreliable-employees

Russell, C. (2025). *No smoke and mirrors: What the adoption of AI in physical security really means.* Keyholding. https://www.keyholding.com/no-smoke-and-mirrors-what-the-adoption-of-ai-in-physical-security-really-means/

Santiago, D. (2025). *Access control: 4,000 years of strategy and innovation + actionable insights for today's physical security leaders.* Site Owl. https://getsiteowl.com/access-control-4000-years-of-strategy-and-innovation-actionable-insights-for-todays-physical-security-leaders/

Williams, C. (2003). Police surveillance and the emergence of CCTV in the 1960s. *Crime Prevention and Community Safety*. *5*(1), 27–37 https://doi.org/10.1057/palgrave.cpcs.8140153